

Lecture 4

Thursday January 29th 2026

OS Structures

Processes and Threads

- A **Process** is a program in execution
 - They will be assigned a process ID that has many uses
- The process can be interrupted to let the CPU execute a higher-priority process
- Need **scheduling** to run the programs and switch between them

Multi-threading

- The execution part is a "thread" that allows multitasking
- Multitasking gives the illusion of parallel processing on one CPU
- The OS is responsible for managing processes and Threads

CPU Scheduling

- As a process executes, it changes state
 - **new:** process is being created
 - **ready:** waiting to be assigned to a processor
 - **running:** instructions being executed
 - **waiting:** waiting for some event to occur
 - **terminated:** the process has finished execution

I/O Management

- Any thing that provides for input and output
 - monitor, printer, mouse, keyboard, speakers
- The devices are all controlled by drivers handled by the OS
 - The drivers tell the OS how to talk to the different devices
- Synchronous or Asynchronous

Memory Management

- OS must fit multiple processes in memory
- Memory needs to be subdivided to accommodate multiple processes

- Memory needs to be allocated to ensure a reasonable supply of ready processes so that the CPU is never idle
- Memory management is an **optimization** task under **constraints**
- fitting processes into memory is like loading boxes into a truck

Examples

- Watch the video to follow along
- CPU utilization = CPU Time / total time
- Want to keep the CPU utilization high to maximize the resource usage
- Efficiency is key to any operation

Memory Storage

- Memory is organized in a hierarchy
- CPU has registers which are the fastest as they are built into the CPU
 - x64 usually has 16 registers at 64 bit
- L1, L2, L3 cache are next fastest but very expensive
- RAM is the working memory
 - dynamic memory non-persistent
- Disk is largest and slowest
 - also called static memory as it is persistent

Read the slides for more information on OS system Design